

ASH AVIATION STORES

Inventory, GRN & Costing Management System

A working digital replacement for the current Excel booking-in, costing and print workflow.

Document	Complete system flow and module guide
Prepared for	Ash Aviation Stores (United Kingdom)
Purpose	Demo, training and approval of the digital stores panel
Currency / dates	GBP (£) · DD/MM/YYYY
Demo login	admin@demo.local · admin123
Date	September 2026
Status	Working demonstration — local data, production-ready product flow

1. Purpose of this document

This guide explains, in plain English, how the new stores system works from start to finish. It is written for managers and stores staff who currently use Excel. It can be used as a presentation script, a training handout, or a record of what has been built for approval.

The software is a **working application**, not a set of static screens. Staff can book goods in, update stock, cost a GRN, print paperwork, search parts and produce reports. Data is saved in the browser so it remains after refresh. After approval, the same screens can be connected to a live company database without rebuilding the front end.

2. The Excel process today

Stores currently run three related spreadsheets:

Excel sheet	What staff do	Typical difficulty
GRN Booking In	Record a delivery: GRN number, supplier, invoice, part number, serial, quantity, store, job	Re-typing part names; mixed spellings; one delivery spread over many rows
GRN Costing Input	Enter cost, calculate sale price and profit	Hidden formulas; VLOOKUP; POA treated as zero and profit going wrong
Print Sheet	Print the GRN for the file or hangar	Print layout separate from live data; copy/paste drift

Stock on the shelf, job cost (for example JOB 123), supplier spend and “where is this washer?” are answered by scrolling, filtering or memory. There is no reliable audit of who changed what.

3. What has been built instead

One admin panel replaces the three sheets and the surrounding paperwork. The **business rules stay the same** — GRN numbers such as GR10-11, part numbers, serials, alternate P/Ns, workpacks, store boxes, automatic mark-up, manual sale prices, POA and print. The Excel look is not copied; the aviation stores process is.

Kept from Excel

- Real GRN numbers (not a new coded format)
- Header plus line items on one GRN
- Alternate part numbers
- Cost bands and POA
- Jobs / workpacks and aircraft registration
- ASH BOX / Paint Room style locations

Improved in the system

- One place for booking, costing and print
- Stock calculated from movements, not a typed cell
- Search by P/N, STD-8, invoice, serial or job
- Job, supplier and store reports in one click
- Cancel a GRN without losing history
- CSV / PDF export for accounts and meetings

4. End-to-end daily flow

This is the complete stores journey. Use this sequence in a live demonstration.

1. MASTER DATA Part, supplier, store and job exist once in the catalogue.	2. BOOK IN Create a GRN: header for the delivery, items for each part.	3. STOCK Save receives stock automatically. Qty rises in the right box.	4. COST Open costing. Sale price from rules, or manual, or POA.	5. PRINT Print sheet or PDF from the same GRN.	6. REPORT Stock, profit, job, supplier or movements as needed.
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Worked example (demo)

1. A delivery arrives from **Adams** for **JOB 123**.
2. Stores open **GRNs** → **New GRN**, keep or enter the GRN number, choose Adams, invoice, tracking, batch, job and aircraft registration.
3. Under **GRN items** they select part **MS123** (alternate **ALT-123**), quantity 50, purchase cost £20, location **ASH BOX 1**.
4. If more parts are on the same invoice they click **Add item** and enter the next line. One save books the whole delivery.
5. **Save GRN and receive stock** — inventory for that part increases by 50. A RECEIPT movement is stored.
6. They open **Cost this GRN**. Cost £20 sits in the £10–£25 band, so automatic sale is **£27** ($\times 1.35$). Total cost £1,000, total sale £1,350, profit £350.
7. If a special price is agreed they switch the line to **Manual override** (for example £32.67). If the part is too expensive or not yet quoted they choose **POA** — the system will not invent a pound figure.
8. They **save costing**, then **Print / PDF**.
9. Later, hangar needs the parts: Inventory → Adjust → **Issue** against JOB 123. Stock falls. JOB 123 still shows what was received, issued, cost and profit.

One sentence for the meeting: book the delivery once, stock updates itself, sale and profit follow company rules, print and reports use the same numbers.

5. How the records fit together

Record	Plain meaning	Example
Part	What the item is — one catalogue card, used on many GRNs	NAS1149F0463P washer
Alternate P/N	Another name for the same part, not a second stock item	STD-8, AN960-416
Supplier	Who it was bought from	Adams, Airpart
Location	Which box or room it is stored in	ASH BOX 2, Paint Room
Workpack / Job	Which aircraft job it belongs to	JOB 123, aircraft G-ICRM
GRN header	One incoming delivery / invoice	GR26734 from Adams
GRN item	One part line on that delivery	50 washers, serial, batch, box
Stock movement	A receipt, issue, return, adjustment or transfer	+50 RECEIPT, -10 ISSUE
Costing	Cost, sale and profit on a GRN line	Automatic £27 or POA

6. Module guide — every section in the panel

6.1 Login and layout

Staff sign in with email and password. The left menu holds every module. The top bar searches across GRN, part number, alternate P/N, supplier, invoice, serial, batch, workpack and location. Sign out is on the right.

Demo: admin@demo.local / admin123 • open <http://localhost:3000> after npm run dev.

6.2 Dashboard

The home screen. Live totals for GRNs, parts, stock quantity, stock cost value, potential sales, potential profit, suppliers and active jobs. A GRN chart, low or zero stock, and recent receipts, movements and costings. This is the picture that previously took several Excel filters to assemble.

6.3 Parts

The product catalogue. Create a part once; select it on every future GRN. Fields: part number, description, category, unit of measure (EA, BOX, LTR, KIT, SET, M, KG), default cost and sale (hints only), status, notes and alternate numbers.

Alternate P/N: the same physical part often has a manufacturer number, an equivalent (AN/NAS), a supplier code and a shop short code. Example: NAS1149F0463P also searched as STD-8. Alternates are search aliases, not extra stock.

The part page shows current stock, last supplier, last cost, movement history, GRN history and linked jobs/costing. Categories include fasteners, washers, bolts, filters, fluids, ignition, seals, hardware, consumables and other.

6.4 Suppliers

Vendor master used on GRNs. Name, contact, email, phone, address, notes, active/inactive. The supplier page shows GRN count, total quantity, purchase value, recent receipts and parts supplied. Demo suppliers: Adams and Airpart.

6.5 Stores / Locations

Physical places in stores — not jobs. Demo locations: ASH BOX 1 (fasteners), ASH BOX 2 (washers/hardware), ASH BOX 3 (overflow), Paint Room (fluids and consumables). Each has a name, short code, description and status. New racks are added here, then chosen on GRN lines. Inventory and the locations report show stock by box.

6.6 Workpacks / Jobs

The aircraft job card. Workpack and job mean the same thing in this system. Number (JOB 123), title, aircraft registration, status (open, in progress, completed, cancelled, archived), dates and notes.

Opening a job answers: which parts were associated, which GRNs, received vs issued quantity, cost and profit. A GRN header can belong to one job; individual lines can belong to different jobs on the same invoice. Demo jobs include JOB 10, 47, 80, 123 and SAM6791.

6.7 GRNs — booking in

This is Excel's Booking In sheet. **All GRNs** lists number, date, supplier, workpack and status (draft, received, cancelled, archived). Search by GRN, invoice, tracking or batch.

New GRN has two blocks:

Header (once per delivery): GR number, date, supplier, invoice, tracking, batch, workpack, registration, sheet number of total sheets, notes, optional document.

GRN items (one row per part): part number, alternate P/N, serial, description, quantity, purchase cost each, batch, store location, workpack, notes. Item 1 is required. **Add item** adds the next Excel-style row on the same GRN. The last item cannot be removed, because an empty GRN is not allowed.

Save receives stock and writes RECEIPT movements. Historical GRNs stay separate even if the same P/N arrives again. **Cancel** reverses the receipt quantities; the GRN is not permanently deleted. Print and PDF are available from the GRN. GRN numbers remain business numbers (GR10-11, GR259-98). Settings can suggest the next sequence; staff can still type the exact Excel number.

6.8 Inventory and stock movements

Current quantity is the sum of movements — it is not a cell staff overwrite. The stock list shows P/N, alternates, quantity, cost, stock value, sale, potential sale, potential profit, location and status. Filter by store. POA lines show POA, not a false pound profit.

Adjust records: Issue (to a job/hangar), Adjustment in, Adjustment out, Return, Transfer. **Stock movements** is the diary: date, part, type, quantity change, GRN, workpack, notes.

6.9 Costing

Excel's Costing Input sheet. Choose a GRN; lines load automatically (no retyping, no VLOOKUP). Four totals: total cost, total sale, profit, profit %.

On each line, cost each comes from the GRN. Sale is set by pricing method:

Method	Who sets the sale price	When to use
Automatic	Company pricing rules (cost × multiplier). Sale field locked.	Normal day-to-day selling
Manual override	Staff type the sale each for this line only	Special deal, exception, agreed job price
POA	No pound sale or profit is shown	Not yet quoted, rare, or high-value band

Figures: total cost = qty × cost each; total sale = qty × sale each; profit = sale – cost; profit % = profit ÷ cost. POA lines are excluded from pound totals so profit cannot be corrupted by a blank or zero.

6.10 Pricing rules

The mark-up bands that used to live in spreadsheet formulas. Each rule has priority, label, minimum cost, maximum cost, multiplier or POA, and active flag. Demo bands:

Purchase cost (each)	Sale price
Under £1	× 3.00
£1 to under £5	× 1.75
£5 to under £10	× 1.45
£10 to under £25	× 1.35
£25 to under £50	× 1.30
£50 to under £100	× 1.20
£100 to under £250	× 1.18
£250 to under £500	× 1.15
£500 and above	POA

Low-cost hardware carries a higher multiple; expensive items a lower multiple or quote. Rules can be edited when company policy changes.

6.11 Reports — complete list

Reports do not create new figures. They cut the live GRN, stock and costing data. Every report can be filtered (dates, search, supplier, part, workpack) and exported to **CSV** (open in Excel) or **PDF**.

#	Report	Question it answers	Main columns
1	Inventory	What is on the shelf now, and what is it worth?	P/N, alternate, qty, cost, stock value, sale, potential sale/profit, location
2	GRNs	What was received, from whom, at what purchase value?	GRN, date, supplier, items, qty, purchase value, workpack, status
3	Costing	Cost, sale and profit on each line?	GRN, P/N, qty, cost each, totals, sale, profit, %, method
4	Profit	Overall profit for a period, supplier or job?	Same lines plus total cost, sale, profit and profit %
5	Suppliers	How much business with Adams vs Airpart?	Name, GRN count, quantity, purchase value (cancelled GRNs excluded)
6	Workpacks	What did JOB 123 consume and make?	Job, title, registration, items, qty, cost, sale, profit
7	Stock movements	When did stock move in or out?	Date, P/N, type, qty, GRN, workpack, notes
8	Locations	What is sitting in each box or room?	Store, code, part count, qty, stock value

6.12 Documents

Attach invoices, delivery notes, certificates, photos or other files to a GRN, part, workpack or supplier. Download or remove from the documents list. A file can also be attached when saving a GRN. This demo stores files locally; a later production step can move them to company cloud storage.

6.13 Excel import

Bring historic workbooks across. Upload .xlsx, .xls or .csv; choose the sheet; map columns (GRN, date, supplier, invoice, tracking, batch, part number, alternate, serial, description, quantity, cost, stores, workpack, notes, sale price); preview valid vs invalid rows; import. Invalid rows are not imported silently. Existing GRN numbers can be preserved.

6.14 Activity log

Who did what: creates, updates, stock movements, costing, imports and report exports. Date/time, user, action, entity, reference, description. Operational records are cancelled or archived rather than hard-deleted, which suits aviation traceability.

6.15 Settings

Company name, address, phone and email (used on print/PDF). GRN prefix, next sequence number, and whether to suggest automatic numbers (still editable on the GRN). **Reset demo data** (with confirmation) reloads the sample aviation dataset in this browser only — useful before a second demonstration, not for live company data.

7. Print sheet and PDF

This is Excel's third sheet. From a GRN choose Print (screen) or download PDF. The sheet uses the live header, line items and costing where a numeric price exists. Company details come from Settings. Staff no longer maintain a separate print workbook.

8. Suggested live demonstration (about 8–10 minutes)

1. Sign in → Dashboard (live totals).
2. Parts: show a washer and alternate STD-8.
3. New GRN: Adams, JOB 123, one or two items, ASH BOX, save — stock rises.
4. Cost the GRN: automatic £27 example; optional manual or POA.
5. Print / PDF.
6. Header search: STD-8 or invoice number.
7. Open JOB 123 — parts, GRNs, cost/profit.
8. Reports → Profit or Workpacks; export CSV or PDF.
9. Refresh the browser — the GRN is still there.

9. Excel compared with the new system

Excel today	Digital system
Three sheets that can drift apart	One panel; one set of numbers
Part names typed on every row	Part chosen from the catalogue
Store names typed freely	ASH BOX / Paint Room from a list
Job filter in a column	Workpack page and workpack report
Stock as a quantity cell	Stock from receipts, issues and adjustments
Formulas and VLOOKUP	Pricing engine + costing workspace
POA or blank can break profit	POA shows POA; no false £ total
Separate print workbook	Print/PDF from the GRN
Pivot tables for reports	Eight ready reports, CSV and PDF
Little history of changes	Activity log; cancel rather than delete
File on one laptop	Application ready to move to a shared company database

10. Benefits for the company

- **Time:** book-in, cost, print and report in one flow instead of three files.
- **Fewer mistakes:** standard mark-up; POA cannot produce a bogus profit.
- **True stock:** goods in raise quantity; issues reduce it; boxes are named.
- **Job costing:** JOB 123 shows parts, cost and profit without rebuilding a pivot.
- **Traceability:** GRN, serial, batch, certificates, who received, who cancelled.
- **History:** existing Excel can be imported; GRN numbers stay as staff know them.
- **Handover:** new staff follow screens rather than undocumented spreadsheet habits.

11. What this demonstration includes — and what comes next

Included now	After approval
• Full stores flow as described above • Sample aviation parts, GRNs, jobs and pricing • Data saved in this browser (survives refresh) • UK money and date formats	• Shared company database (for example PostgreSQL / Supabase) • Real user accounts, backups and hosting • Optional cloud document storage • The same screens — not a rebuild

This demo runs on one computer's browser. Two people on two PCs do not yet share a live database. That is the intended next production step.

12. Sample data loaded for the demo

- Company: Ash Aviation Stores, Hangar 4, Ashford Aerodrome, Kent
- Suppliers: Adams, Airpart
- Locations: ASH BOX 1–3, Paint Room
- Jobs: JOB 10, 47, 80, 123, SAM6791
- Parts including NAS washers (STD-8), filters, spark plugs, MS123 / ALT-123
- Pricing tiers as in section 6.10
- Historic sample GRNs so reports are not empty; next suggested number around GR26734

13. Closing statement for the presentation

Ash Aviation Stores currently books goods in, costs them and prints from three Excel sheets. This system keeps those same aviation fields and pricing habits, and puts them in one working panel: GRN header and items, live stock, automatic or manual or POA selling prices, print, documents, Excel import, job and supplier views, and eight reports. The demonstration is intended for approval of the process. A shared company database can follow without changing how staff already use the screens.